

ENGINEERING BULLETIN

Product Options

Siemens Software New Revision Updates

No: EB/23-24/001,
10/08/2023

Dear Colleague,

We here by wish to inform you about the official release of Siemens controller software revision with new updates.

Software Version: This manual covers units with Software Version for Single & Dual circuit, the unit's software version number can be viewed by selecting the Software Details (Inside Parameters) for Keypad display & for Touch screen on main overview page right side bottom, both item accessible without password. Then, pressing the left side top key will return to the Menu screen for keypad display only.

Minimum BSP Version: 11.48

Siemens Single Circuit Software Version – SG.00.07.07

Siemens Dual Circuit Software Version – SG.00.07.07

Below updated points for single circuit software.

- 1 HMI status correction done for all digital input/output.
- 2 Main Screen Timer for Compressor Anti cycle, EXV Startup Time and Oil Feed SoV are provided on display.
- 3 Desuperheater water temperature (in/out) showing on condenser display.
- 4 Desuperheater heater enable/disable option provided on display.
- 5 Water flow meters display on the EVAP and COND pages.
- 6 Pressure probe fault all alarms tested on touch screen.
- 7 Low demand notification showing on Main screen instead of READY notification (so during auto off on LWT setpoint, before shows READY)
- 8 Oil and Desuperheater water temp (in/out) offset function is provided on display.
- 9 All Digital Input alarms are checked and provided on display.
- 10 Compressor Unload status showing on the compressor display (current/SP/DP/).
- 11 PID logic page show on setpoint screen and few setpoint working provided on display.
- 12 Previous Run hours and number of start addition logic added and provided on display.
- 13 LWT control band (Compressor shutdown modifications logic) logic modified and checked.
- 14 Low Load Auto OFF Notification is shows on alarm history page.
- 15 Touch screen – Clear Log data Key removed.
- 16 Touch screen - Log data download in user password is changed (Old User password 4321 & now 1234).
- 17 Password Log out KEY developed & provided on display (Password disable function).
- 18 Password auto-disable during controller power cycle.

- 19 EXV function Safety unloads on Suction pressure is done.
- 20 Fault and indication of VPD/starter fault/SS is done
- 21 EXV open/close status update on HMI EXV screen (SSH/SP /Differential Oil PR)
- 22 VFD current setpoint 1 is provided on display.
- 23 Oil level switch timer setting is provided on display.
- 24 Oil level trip fault indication is provided on display
- 25 Working of VFD current Provided on display.
- 26 Comp 1 Anit cycle timer is Done
- 27 Desuperheater water temp (in/OUT) showing on condenser display.
- 28 Key functions of Desuper heater enable/disable is checked
- 29 Alarm History is tested
- 30 Compressor Unload status showing on the compressor display (current/SP/DP/)
- 31 Probe offset function is checked.
- 33 Trip/Alarm Relay Contact (Relay ON/OFF) checked.
- 34 Chiller ON/OFF testing is done, and the same program is uploaded in the second chiller unit.
- 35 Chiller wiring was changed as per the latest wiring diagram, and Trip/Alarm Relay Contact (Relay ON/OFF).
- 36 Cooling tower fan ON/OFF is also provided.

Below updated points for dual circuit software.

1. Previous Run hours and no start addition logic added & also provided on display.
2. LWT control band (Compressor shutdown modifications logic) logic modified and checked and working provided on display as per our requirement.
3. Touch screen: log data disable key removed
4. Touch screen: Log data download in user password changed (Old User password 4321 and now 1234)
5. Password auto-disable during controller power cycle: found Working provided on display.
6. EXV function Safety unloads on Suction pressure.
7. SP trip function and alarm are provided on display.
8. Antifreeze trip function and Alarm are provided on display.
9. Discharge Temp trip fault and indication provided on display.
10. Oil temp trip fault and indication provided on display
11. VPD / starter fault/SS--1 & 2 fault and Indication provided on display.
12. EXV open/close status update on HMI EXV screen (SSH/SP /Differential Oil PR)
13. VFD current set points 1 and 2 are provided on display.
14. Oil level switch timer setting is provided on display.
15. Oil level trip fault indication is provided and shows on display.
16. VFD current function Working is Provided on display.
17. Desuperheater water temp (in/OUT) showing on condenser display.
18. Desuperheater heater enable/disable -- Key function checked and working provided on display
19. Alarm History is provided on display.
20. Compressor Unload status added on the compressor display
21. Timer display added on HMI for Compressor Antirecycle, EXV Startup Time, Oil

Feed

22. Demand set point changing function on HMI.
23. EXV 1 and 2 display graphics are provided on display.
24. Compressor Loading / Unloading - Parallel Operation (Load / Unload Band).
25. As per Carel, Compressor Unload Status shows on Compressor Menu.
26. PID SP on the HMI set point screen and working provided on display.
27. All Sensors and Pressure Probe Fault Alarm are Checked and its working provided on display.
28. Comp 1 & 2 Low load auto off Notification on the Alarm menu.
29. Data log on HMI.

Control System Diagnostic:

POL 688 (K-smart) controller, extension modules and communication modules are equipped with two statuses LED (BSP and BUS) to indicate the operational status of the devices. The meaning of the two statuses LED is indicated below.

- **Controller LED**

BSP LED	BUS LED	Mode
Solid Green	OFF	Application running
Solid Yellow	OFF	Application loaded but not running (*)
Solid Red	OFF	Hardware Error (*)
Flashing Yellow	OFF	Application not loaded (*)
Flashing Red	OFF	BSP Error (*)
Flashing Red/Green	OFF	Application/BSP update

(*) Contact your immediate officer.

- **Extension Module LED**

BSP LED	BUS LED	Mode
Solid Green		BSP running
Solid Red		Hardware Error (*)
Flashing Red		BSP Error (*)
	Solid Green	Communication running, I/O working
	Solid Yellow	Communication running, parameter missing (*)
	Solid Red	Communication down (*)

(*) Contact your immediate officer.

PID settings options available on all HMI's for various tuning purpose.

- Proportional stand for control factor
- Integral stand for action time
- N Band stand for Neutral Band

Proportional time close to zero & Integral time at higher side means PID outputs vary slowly.

POL HMI PID Set Points Screen.

RLA Ctrl	100.0%
RLA Ctrl P	-2
RLA Ctrl Intgrl	300s
Disch Pr Ctrl SP	8.0bar
Disch P N Bnd	0.2bar
Disch Pr Ctrl Prp.	-10
Disch Pr Ctrl Intgrl	60s
Suc Pr N Bnd	0.5bar
Suc Pr Ctrl Prp.	10
Suc Pr Ctrl Intgrl	30s
Disch Temp Ctrl SP	80.0°C
D Temp N Bnd	5.0°C
Disch Temp Ctrl Prp.	-2
Disch Temp Ctrl Intgrl	300s
Oil Temp Ctrl SP	80.0°C
Oil T N Bnd	5.0°C
Oil Temp Ctrl Prp.	-5
Oil Temp Ctrl Intgrl	90s
EXV SH Ctrl Prp	-15
EXV SH Ctrl Intgrl	20s
EXV Suc Pr Ctrl Prp	10
EXV Suc Pr Ctrl Intgrl	10s
EXV Diff Pr Ctrl Prp	-10
EXV Diff Pr Ctrl Intgrl	20s
EXV DSH Ctrl Prp	-10
EXV DSH Ctrl Intgrl	20s

****Note- Sign for Proportional factor indicates its operational zone during settings/tuning signs (-ve/+ve) should be kept as it is. If changed, then the operation can be reversed.**

- Refer below HMI screen for Dual circuit PID setting's just for reference.

PID SP

: Login Status
21/09/2022 18:50:51

		DISCHARGE CTR				COMP PID CTRL		RLA CTR		RLA VFD CTR		CONTROL SP POWER SP TEMPERATURE SP PRESSURE SP TIMER SP COMPRESSOR SP PROBE SP EXV SP VFD SP		
PROPORTIONAL		-123	-123	-123	-123	-123	-123	-123	-123	-123	-123			
INTEGRAL		-123	-123	-123	-123	-123	-123	-123	-123	-123	-123			
		DSH CTR				EXV PID CTRL		P.SUC CTR		SSH CTR		DIFF CTR		
PROPORTIONAL		-123	-123	-123	-123	-123	-123	-123	-123	-123	-123	-123	-123	
INTEGRAL		-123	-123	-123	-123	-123	-123	-123	-123	-123	-123	-123	-123	

--- NO ALARM ---

COMP 1	##	-123	COMP 2	##	-123
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OVERVIEW

ALARM

DATA LOG

LOGIN

USER SP

SUPERVISOR SP

- Refer below HMI screen for Single circuit PID setting's just for reference.

PID SP

: Login Status
21/09/2022 18:50:51

		EXV PID		PROP NL	INTG RL	CONTROL SP POWER SP TEMPERATURE SP PRESSURE SP TIMER SP COMPRESSOR SP PROBE SP EXV SP VFD SP
Suct Pr. EXV Ctrl SP		-123	-123			
DSH EXV Ctrl SP		-123	-123			
P DIFF EXV Ctrl SP		-123	-123			
SSH EXV Ctrl SP		-123	-123			

COMP PID	SP	NEUTRL BND	PROP NL	INTG RL
Suc Pr. Ctrl SP	-123	-12.3	-123	-123
Disch Pr. Ctrl SP	-123	-12.3	-123	-123
% Load Ctrl SP	-123		-123	-123
VFD RLA Ctrl SP	-123		-123	-123
Disch T. Ctrl SP	-123	-12.3	-123	-123
Oil T. Ctrl SP	-123	-12.3	-123	-123

--- NO ALARM ---

AAAAAAAAAAAAAAAAAAAA 1234

OVERVIEW

ALARM

DATA LOG

LOGIN

USER SP

SUPERVISOR SP